EV Battery Course — Module 17: Advanced Engineering Calculations and Simulation

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This module pulls the course together into a calculation toolkit, from simple pack sizing to a second-by-second drive-cycle simulation that predicts 151 Wh/km and about 387 km of range for the Module 6 pack. Each problem increases in difficulty. Your master prompt was cut off at "charging-time estimation" for this module, so it also covers the natural continuation: thermal transients, equivalent-circuit response, degradation projection and battery cost per km.

1. Learning objectives

- Size packs for two-wheelers and cars from requirements.
- Calculate voltage, current, C-rate, energy and power density at cell and pack level.
- Estimate charging time with taper and losses.
- Solve lumped thermal transients and equivalent-circuit pulse responses.
- Project degradation and battery cost per km.
- Build and interpret a simple drive-cycle simulation in Python.

2. Fundamentals: the modelling ladder

Level	Model	Use
1	Hand equations (E = VC, P = VI, I^2R)	Concept sizing, sanity checks
2	Spreadsheet with lookup tables (OCV-SOC, R vs T)	Pack trade studies, costing
3	Time-step simulation (vehicle + ECM + thermal)	Range, power, thermal over drive cycles
4	Electrochemical models (P2D, SPMe)	Cell design, ageing, fast-charge limits
5	3D multiphysics (CFD, FEA)	Thermal, mechanical and crash detail

Always sanity-check a higher-level result against a lower-level estimate.

3. Formula sheet

Quantity	Formula
Cell energy	$E = V_{nom} \times C$
Pack voltage / capacity	$V = N_s V_{cell}$; $C = N_p C_{cell}$
Current	$I = P \div V$
C-rate	$I \div C_{pack}$
Specific energy / power	$E \div m$; $P \div m$
Loss / sag	I^2R; IR
Current for power with sag	$I = [V_{oc} - \sqrt{(V_{oc}^2 - 4RP)}] \div (2R)$
Road load	$F = ma + mgC_{rr} + \frac{1}{2}\rho C_d A v^2$ (+ $mg \sin\theta$ on grades)
Charge time	$t = \Sigma \Delta E_i \div (P_i \eta_i)$
Lumped thermal	$m c_p dT/dt = Q - hA(T - T_c)$
RC pulse	$V = OCV - IR_0 - IR_1(1 - e^{(-t/\tau)})$
Calendar + cycle fade	$Q_{loss} = k\sqrt{t} + b \times EFC$

The current-with-sag formula comes from $P = I(V_{oc} - IR)$, a quadratic in I: $RI^2 - V_{oc} I + P = 0$.

4. Worked problems (increasing difficulty)

Problem 1 — Two-wheeler pack sizing

Requirement: 100 km real range at 30 Wh/km; 90% usable; peak 5 kW; cells 21700 NMC, 3.6 V, 5 Ah (18 Wh), max 3C pulse.

1. Usable energy = 100 × 30 = 3.0 kWh → gross = 3.0 ÷ 0.9 = **3.33 kWh**.
2. Choose 16S for ~58 V: 16 × 3.6 = **57.6 V** (stays near the 60 V safety threshold at nominal; check maximum 16 × 4.2 = 67.2 V).
3. Ah needed = 3,333 ÷ 57.6 = 57.9 Ah → **12P** (60 Ah).
4. Pack **16S12P = 192 cells = 3.46 kWh**.
5. Peak current = 5,000 ÷ 57.6 = 86.8 A → per cell 7.2 A = **1.45C**: within the 3C pulse limit.

Problem 2 — Power density

The Module 6 pack (468 kg) delivers 150 kW peak: 150,000 ÷ 468 = **320 W/kg** at pack level, versus 131 Wh/kg energy density. Ratio P/E = 150 ÷ 61.44 = 2.4 (a power-to-energy ratio of about 2.4 h^{-1}, i.e. 2.4C).

Problem 3 — Charging time with taper

See Module 9: 10–80% on a 100 kW charger = 15.5 + 11.1 + 8.6 = **35.2 min** using three SOC bands and 95% efficiency.

Problem 4 — Thermal transient with cooling

Pack heat Q = 3.5 kW (fast charge); thermal mass mc_p = 351 kJ/K; cooling conductance hA = 350 W/K (cells to coolant).

- Steady-state rise above coolant: ΔT_ss = Q ÷ hA = **10 K**.
- Time constant τ = mc_p ÷ hA = 351,000 ÷ 350 = **1,003 s (16.7 min)**.
- After 30 min: ΔT = 10 × (1 − e^(−1,800/1,003)) = **8.3 K** above coolant.

Problem 5 — Equivalent-circuit pulse response

Cell: R_0 = 0.25 mΩ, R_1 = 0.15 mΩ, τ_1 = 10 s (C_1 = τ ÷ R_1 = 66,667 F). 200 A pulse for 10 s.

- Instant drop: 200 × 0.00025 = 50 mV.
- RC drop after 10 s: 200 × 0.00015 × (1 − e^{-1}) = 19 mV.
- Total = **69 mV**. A DCIR measured at 10 s would read 0.069 ÷ 200 = 0.345 mΩ, higher than R_0: the pulse duration must always be stated with DCIR.

Problem 6 — Degradation projection

Model: Q_loss = 2.5% × √(years) + 0.004% per EFC. Private car: 35 EFC/year for 8 years (280 EFC).

- Calendar: 2.5 × √8 = 7.07%. Cycle: 0.004 × 280 = 1.12%. Total **8.2%** → SOH ≈ **91.8%**: comfortable against a 70% warranty threshold. (Model coefficients are illustrative; fit them to your cell's test data.)

Problem 7 — Battery cost per km

Pack $4,915 (Module 6); assume it delivers 2,000 full equivalent cycles × 58.4 kWh before end of life in a high-use fleet.

- Cost per kWh of throughput = 4,915 ÷ (2,000 × 58.4) = **$0.042/kWh**.

- At 0.151 kWh/km: **$0.0064/km** (about ₹0.56/km at ₹88/$). Add electricity cost for full energy cost per km. This is the logic behind fleet TCO comparisons.

5. Simulation project: drive-cycle range model

The script below simulates the Module 6 vehicle (1,700 kg, 120S LFP pack) over a synthetic cycle of 10 urban stop-go blocks (0→50→0 km/h) followed by a highway run (0→90 km/h, 5 minutes cruise, stop). Every second it computes acceleration and road-load force, wheel power, battery power (with drivetrain losses, regen efficiency and auxiliaries), battery current including voltage sag, energy, losses and SOC.

```
import math
m, g, crr, cda, rho = 1700, 9.81, 0.009, 0.32*2.4, 1.2   # vehicle data
eta_d, eta_r, aux = 0.87, 0.65, 600.0                     # drivetrain eff.,
regen eff., aux power W
Voc, R, C_ah = 384.0, 0.040, 160.0                         # pack OCV,
resistance, capacity

# speed profile in km/h, 1-second steps
v = []
for _ in range(10):                       # 10 urban stop-go blocks
    v += [5*i for i in range(1, 11)]      # accelerate 0->50 in 10 s
    v += [50]*30                          # cruise 30 s
    v += [50-5*i for i in range(1, 11)]   # brake 50->0 in 10 s
    v += [0]*10                           # stand 10 s
v += [4.5*i for i in range(1, 21)]        # highway 0->90 in 20 s
v += [90]*300                             # cruise 5 min
v += [90-4.5*i for i in range(1, 21)]     # brake to stop

soc, dist, e_batt, e_loss, i_max, prev = 1.0, 0.0, 0.0, 0.0, 0.0, 0.0
for kmh in v:
    s = kmh/3.6; a = s - prev/3.6; prev = kmh               # speed m/s,
accel m/s^2
    F = m*a + (m*g*crr + 0.5*rho*cda*s*s if s > 0 else 0)  # inertia +
rolling + aero
    Pw = F*s                                                # wheel power, W
    Pb = (Pw/eta_d if Pw >= 0 else Pw*eta_r) + aux          # battery power,
W
    I = (Voc - math.sqrt(Voc*Voc - 4*R*Pb))/(2*R)           # solve P = I(Voc
- IR)
    i_max = max(i_max, I)
    e_batt += Voc*I/3600; e_loss += I*I*R/3600              # Wh from cells,
Wh lost in R
    soc -= I/3600/C_ah; dist += s/1000                      # SOC and km
```

```
print(f"time {len(v)} s, distance {dist:.2f} km")
print(f"energy from cells {e_batt:.0f} Wh -> {e_batt/dist:.0f} Wh/km")
print(f"resistive loss {e_loss:.1f} Wh, peak current {i_max:.0f} A, SOC used
{(1-soc)*100:.2f}%")
```

Output when run:

Result	Value
Duration / distance	940 s / 13.56 km
Energy from cells	2,041 Wh → **151 Wh/km**
Resistive loss in pack	13.5 Wh (0.7%)
Peak current	197 A (about 1.2C)
SOC used	3.32%

Interpretation. 151 Wh/km matches the hand estimate of 145–147 Wh/km from Module 6, adding confidence. Implied range with 58.4 kWh usable: 58,368 ÷ 151 ≈ **387 km**. Pack resistive loss is small in normal driving; it matters at peak power and fast charging.

Extend it yourself:

1. Replace the synthetic cycle with the real MIDC or WLTC speed trace (from published tables).
2. Make Voc a function of SOC (OCV table) and R a function of temperature.
3. Add a lumped thermal model (Problem 4) and watch temperature rise during a hill climb.
4. Add a road grade (mg sinθ) and compare Ghat climbs with flat cities.
5. Plot speed, current and SOC over time with matplotlib.

6. Real-world use

- OEMs use tools such as MATLAB/Simulink, GT-SUITE, AVL Cruise, and Python models at levels 2–3; cell makers and research labs use PyBaMM (open-source Python) or COMSOL at level 4.
- Homologation range (MIDC, WLTP) and real-world range differ mainly because of speed, temperature, HVAC load and driver behaviour, all of which a simulation can include.

7. Manufacturing and costing applications

- Use Problem 7's logic to compare chemistries on cost per km, not just cost per kWh.
- Use simulation outputs (peak current, RMS current, heat) to size busbars, fuses and cooling, which then feed BOM costs.
- Tolerance analysis: run the model with cell capacity and resistance at ±3σ to see the effect of manufacturing spread on range and power.

8. Common mistakes

- Mixing kW and kWh, Ah and Wh, or km/h and m/s.
- Forgetting auxiliaries (HVAC can add 1–3 kW in Indian summers).
- Treating regen as 100% efficient.
- Using nominal voltage at peak power (sag raises current).
- Not validating the model against test data.

9. Practical case study: range shortfall in validation

Problem. Prototype tests show 330 km against a simulated 387 km.

Investigation. Model used 600 W auxiliaries; measured average AC load was 1.8 kW at 40 °C ambient. Tyre rolling resistance on the prototype was 0.011, not 0.009.

Correction. Update model inputs; rerun: about 335 km, matching test. Engineering actions: low-rolling-resistance tyres and a heat pump option.

Lesson. A simulation is only as good as its inputs; validate and update.

10. Revision notes

- Size: range × Wh/km ÷ usable window; choose N_s for voltage, N_p for Ah.
- Check C-rate at peak and continuous power; use the sag-corrected current.
- Thermal: $\Delta T_{ss} = Q/hA$; $\tau = mc_p/hA$.
- ECM: instant R_0 drop plus RC relaxation; quote DCIR with pulse time.
- Validate simulations against hand estimates and tests.

11. Glossary

Term	Meaning
ECM	Equivalent-circuit model
P2D / SPMe	Electrochemical cell models
PyBaMM	Open-source Python battery modelling library
RMS current	Root-mean-square current; drives heating
Drive cycle	Standard speed-vs-time profile for testing
Time constant (τ)	Time to reach ~63% of a step response

12. Knowledge check

Conceptual

1. Why must DCIR be quoted with a pulse duration?
2. Why does voltage sag increase current for a given power?
3. What is the physical meaning of the thermal time constant?
4. Why can a simulation and a homologation test disagree?
5. Why compare chemistries on cost per km?

Numerical

1. An e-rickshaw needs 120 km at 45 Wh/km, 90% usable, 51.2 V LFP (16S). Gross kWh and cell Ah?
2. Solve for current: V_{oc} = 400 V, R = 0.05 Ω, P = 120 kW.
3. Q = 2 kW, hA = 250 W/K, mc_p = 200 kJ/K. Find ΔT_{ss}, τ and ΔT after 20 min.

Application

1. Extend the simulation to estimate range at 100 km/h steady speed. What do you expect and why?
2. How would you use simulation to set the charging-curve limits for a new cell?

13. Answer key

Conceptual

1. Voltage keeps dropping after the instant ohmic step because of charge-transfer and diffusion processes; longer pulses give higher apparent resistance.

2. Terminal voltage falls under load (V = V_oc − IR), so more current is needed for the same V × I.
3. It is the time the pack takes to reach about 63% of its final temperature rise after a step heat load; it compares thermal mass with cooling strength.
4. Different speed profiles, temperatures, HVAC loads, test procedures and vehicle parameters.
5. Life (cycles) and cost together decide the cost of each kilometre; a cheap chemistry with short life can cost more per km.

Numerical

1. 120 × 45 = 5.4 kWh usable ÷ 0.9 = **6.0 kWh**; 6,000 ÷ 51.2 = **117 Ah** (e.g. a 120 Ah LFP cell in 16S1P).
2. I = [400 − √(160,000 − 4 × 0.05 × 120,000)] ÷ (2 × 0.05) = [400 − √136,000] ÷ 0.1 = (400 − 368.8) ÷ 0.1 = **312 A** (versus 300 A without sag).
3. ΔT_ss = 2,000 ÷ 250 = **8 K**; τ = 200,000 ÷ 250 = **800 s**; after 1,200 s: 8 × (1 − e^(−1.5)) = 8 × 0.777 = **6.2 K**.

Application

1. At 100 km/h aero force ≈ 0.5 × 1.2 × 0.768 × 27.8² ≈ 356 N, rolling 150 N → wheel power ≈ 14.1 kW → battery ≈ 16.8 kW incl. aux → about 168 Wh/km → about 348 km. Range falls because aero drag grows with v².
2. Build an electrochemical (P2D) model calibrated with three-electrode cell tests; simulate anode potential during charge at various C-rates, SOCs and temperatures; set current limits so anode potential stays above 0 V vs Li with margin; validate on cells with plating detection; encode as a BMS current map.

14. Further learning

- PyBaMM documentation and examples (pybamm.org).
- Plett, *Battery Management Systems* Vols. I–II (modelling and estimation).
- Ehsani et al., *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles* (vehicle dynamics and powertrain sizing).
- MIDC and WLTC speed traces (ARAI / UNECE publications).

This completes the 17-module course. Suggested next step: rebuild the Module 6 pack in a spreadsheet with your own supplier quotes and run the simulation with a real MIDC trace.